

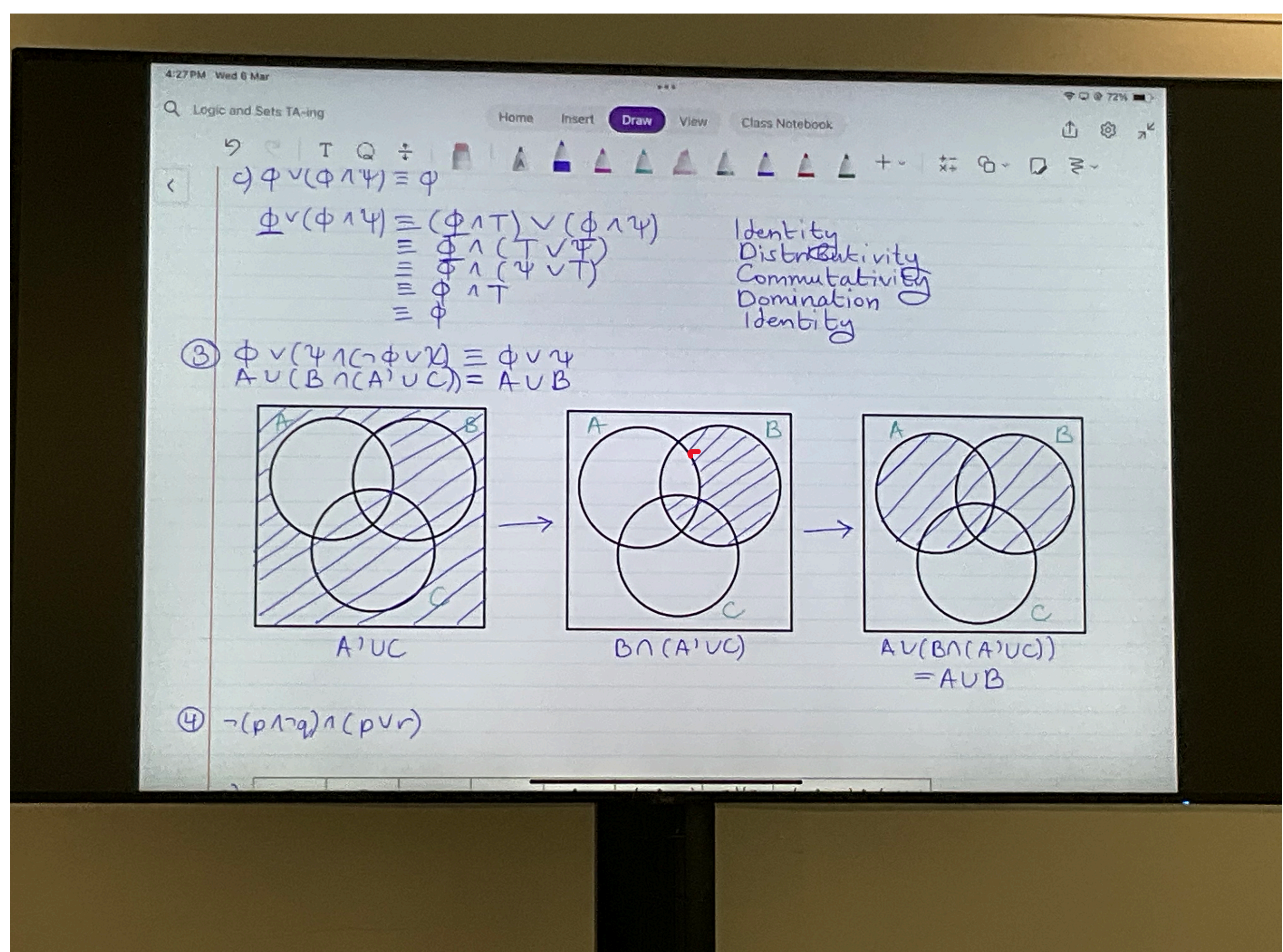
2) a)  $\neg \perp \equiv \top$   
 $\neg \perp \equiv \neg(\phi \wedge \neg \phi)$  complement  
 $\equiv \neg \phi \vee \neg \neg \phi$  De Morgan  
 $\equiv \neg \phi \vee \phi$  involution  
 $\equiv \phi \vee \neg \phi$   
 $\equiv \top$

b)  $(\phi \vee \psi) \vee (\phi \wedge \psi) \equiv \phi \vee \psi$   
 $(\phi \vee \psi) \vee (\phi \wedge \psi) \equiv ((\phi \vee \psi) \vee \phi) \wedge ((\phi \vee \psi) \vee \psi)$  Distributivity  
 $\equiv (\phi \vee \psi) \vee \phi) \wedge (\phi \vee (\psi \vee \psi))$  Associativity  
 $\equiv (\psi \vee (\phi \vee \phi)) \wedge (\phi \vee (\psi \vee \psi))$  Commutativity + Associativity  
 $\equiv (\psi \vee \phi) \wedge (\phi \vee \psi)$  Idempotence x2  
 $\equiv (\phi \vee \psi) \wedge (\phi \vee \psi)$  Commutativity  
 $\equiv \phi \vee \psi$  Idempotence

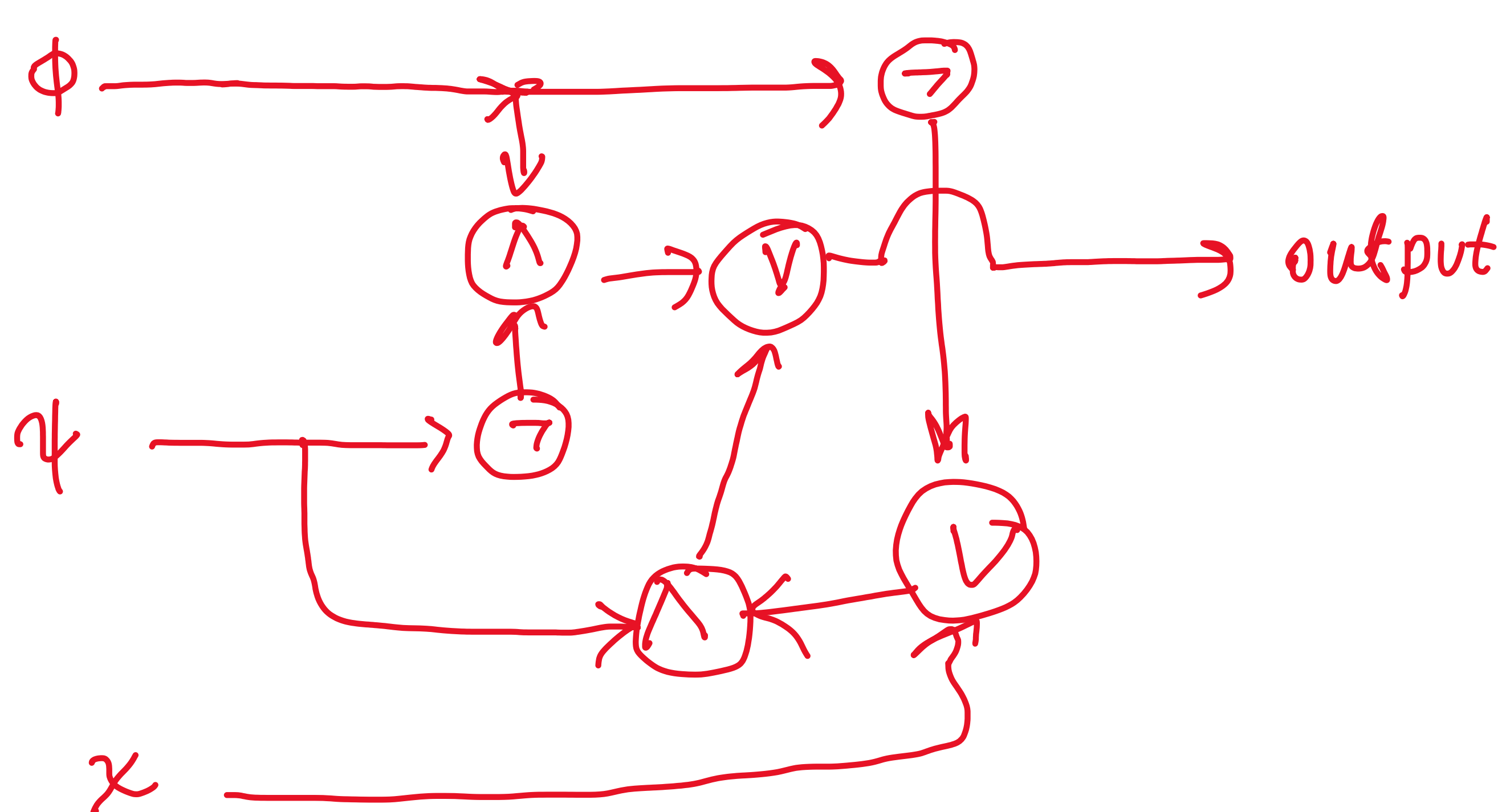
c)  $\phi \vee (\phi \wedge \psi) \equiv \phi$   
 $\phi \vee (\phi \wedge \psi) \equiv (\phi \vee \top) \wedge (\phi \wedge \psi)$  Identity  
 $\equiv \phi \vee (\top \wedge \psi)$  Distributivity  
 $\equiv \phi \wedge (\psi \vee \top)$  Commutativity  
 $\equiv \phi \wedge \top$  Domination  
 $\equiv \phi$  Identity

d)  $\phi \wedge (\phi \vee \psi) \equiv \phi$   
 $\phi \wedge (\phi \vee \psi) \equiv (\phi \vee \perp) \wedge (\phi \vee \psi)$  Identity  
 $\equiv \phi \vee (\perp \wedge \psi)$  Distributivity  
 $\equiv \phi \vee (\psi \wedge \perp)$  Commutativity  
 $\equiv \phi \vee \perp$  Domination  
 $\equiv \phi$  Identity

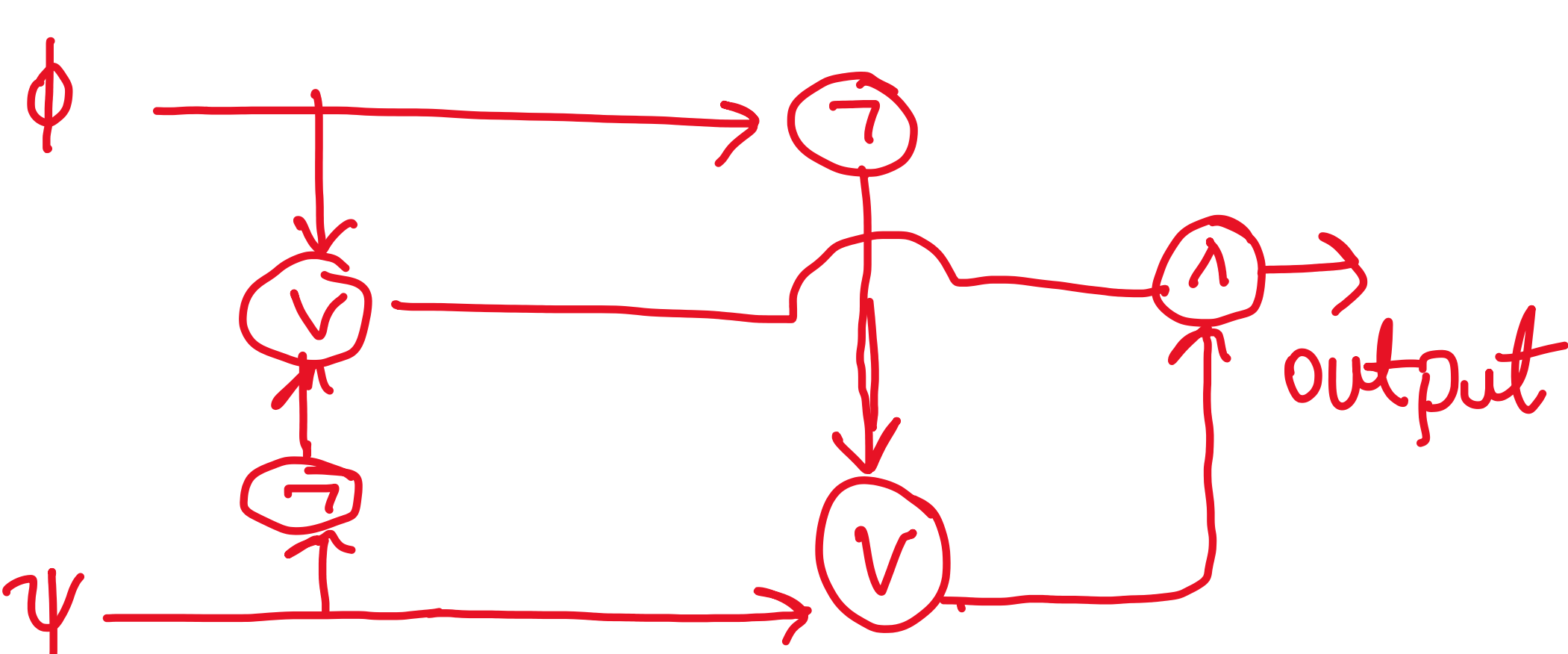
e)  $\phi \vee (\psi \wedge (\neg \phi \vee X)) \equiv \phi \vee \psi$   
 $\phi \vee (\psi \wedge (\neg \phi \vee X)) \equiv (\phi \vee \psi) \wedge (\phi \vee (\neg \phi \vee X))$   
 $\equiv (\phi \vee \psi) \wedge (\phi \vee \neg \phi \vee X)$   
 $\equiv (\phi \vee \psi) \wedge (\top \vee X)$   
 $\equiv (\phi \vee \psi) \wedge (X \vee \top)$   
 $\equiv (\phi \vee \psi) \wedge \top$



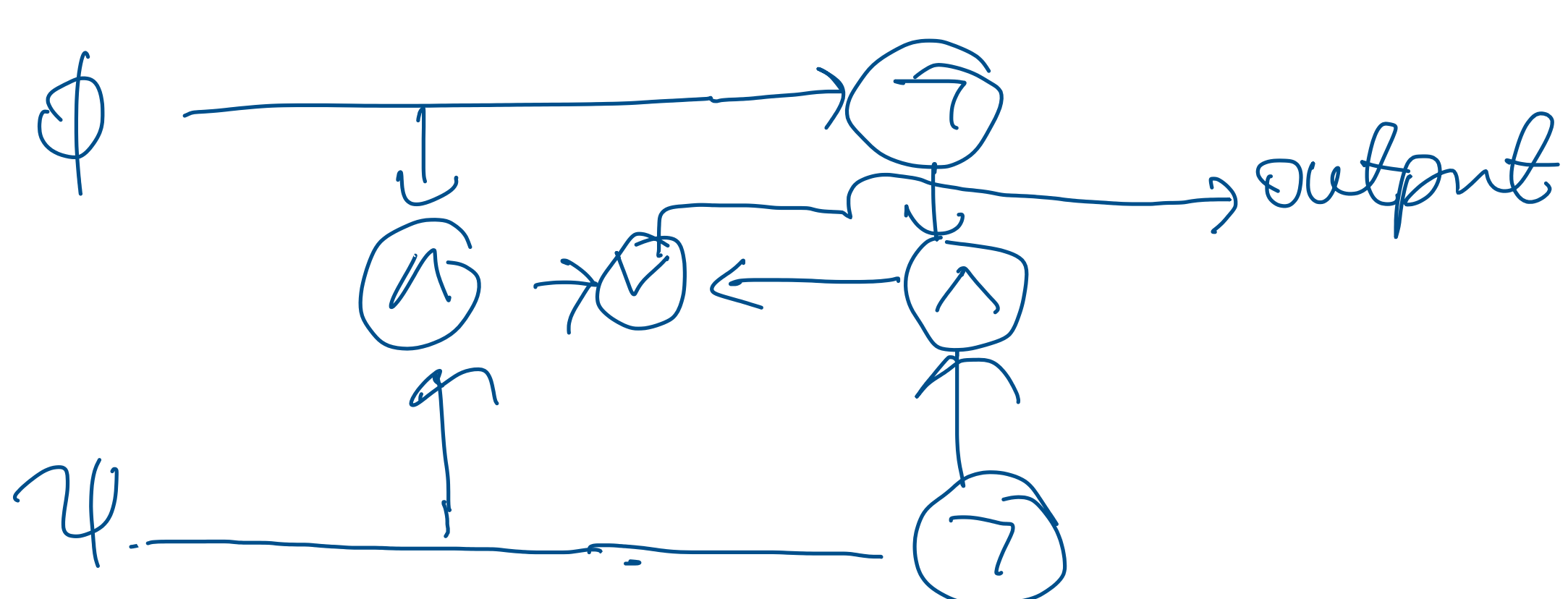
5)  $(\phi \wedge \neg \psi) \vee (\psi \wedge (X \vee \neg \phi))$



6)  $\phi \leftrightarrow \psi \equiv (\phi \rightarrow \psi) \wedge (\psi \rightarrow \phi)$   
 $\equiv (\neg \phi \vee \psi) \wedge (\neg \psi \vee \phi)$



$\phi \leftrightarrow \psi \equiv (\phi \wedge \psi) \vee (\neg \phi \wedge \neg \psi)$



7)  $(\neg x \wedge (\neg y \vee \neg z)) \vee (x \wedge ((y \wedge z) \vee (y \wedge \neg z)))$  min  
 $((y \wedge z) \vee (\neg y \wedge \neg z)) \wedge x) \vee (\neg x \wedge (\neg y \vee \neg z))$  TA

8a) 1101  
 b)  $\begin{array}{r} 100011 \\ + 10110 \\ \hline 1010001 \end{array} \rightarrow 2^8 + 2^1 + 2^0 = 35$   
 $\rightarrow 46$   
 $\rightarrow 81$

9)  $X \wedge \neg X$   
 $X \wedge Y$   

| X | Y | X.Y | P <sub>1</sub> | P <sub>0</sub> |
|---|---|-----|----------------|----------------|
| 1 | 1 | 1   | 0              | 1              |
| 1 | 0 | 0   | 0              | 0              |
| 0 | 1 | 0   | 0              | 0              |
| 0 | 0 | 0   | 0              | 0              |

10)  $\neg(p \wedge q) \wedge (p \vee r)$

| p | q | r | $\neg q$ | $p \wedge \neg q$ | $\neg(p \wedge \neg q)$ | $p \vee r$ | $\neg(p \wedge \neg q) \wedge (p \vee r)$ |
|---|---|---|----------|-------------------|-------------------------|------------|-------------------------------------------|
| T | T | T | F        | F                 | T                       | T          | T                                         |
| T | T | F | F        | F                 | T                       | T          | T                                         |
| T | F | T | T        | T                 | F                       | T          | F                                         |
| T | F | F | T        | T                 | F                       | T          | F                                         |
| F | T | T | F        | F                 | T                       | T          | T                                         |
| F | T | F | F        | F                 | T                       | F          | F                                         |
| F | F | T | T        | F                 | T                       | T          | T                                         |
| F | F | F | T        | F                 | T                       | F          | F                                         |

b)  $(p \wedge q \wedge r) \vee (p \wedge q \wedge \neg r) \vee (\neg p \wedge q \wedge r) \vee (\neg p \wedge q \wedge \neg r)$   
 c)  $\neg(p \wedge q) \wedge (p \vee r) \equiv$

De Morgan  
 Involution  
 Distributivity  
 Complement  
 Commutativity  
 Identity x2  
 Complement  
 Distributivity  
 Idempotence  
 Commutativity